Python Intermediate Day 2 Checkpoint

Object-Oriented Programming Fundamentals
- OOP – The Big Picture
- Class Coding Basics
-Realistic OOP Example
- Class Coding Details

1. Email *

2. Name

3. **What best describes the difference between a class and an instance?** 1 point

Mark only one oval.

- A. A class is data, an instance is code
- B. A class is a blueprint, an instance is a created object
- C. A class exists at runtime, an instance exists at compile time
- D. There is no difference

4. **Which concept allows a subclass to provide its own implementation of a method?** 1 point

Mark only one oval.

- A. Encapsulation
- B. Inheritance
- C. Polymorphism
- D. Composition

5. **Which method is automatically called when an object is created?** 1 point

Mark only one oval.

- A. new
- B. start
- C. init
- D. create

6. **What does the self keyword represent inside a class method?** 1 point

Mark only one oval.

- A. The class itself
- B. The current instance
- C. A global variable
- D. A parent class

7. **Which method should be overridden to provide a readable string for users?** 1 point

Mark only one oval.

- A. repr
- B. debug
- C. str
- D. print

8. **What is the main purpose of inheritance?** 1 point

Mark only one oval.

- A. To reduce memory usage
- B. To reuse and extend existing code
- C. To hide data
- D. To prevent modification

9. **What does super() do?** 1 point

Mark only one oval.

- A. Calls a static method
- B. Refers to the current object
- C. Calls methods from the parent class
- D. Creates a new instance

10. **Which tool allows inspection of object attributes at runtime?** 1 point

Mark only one oval.

- A. print()
- B. help()
- C. introspection (e.g., dir(), hasattr())
- D. compile()

11. **Which module is commonly used for object persistence in Python?** 1 point

Mark only one oval.

- A. json
- B. csv
- C. pickle
- D. sys

12. **A lambda function in Python is:** 1 point

Mark only one oval.

- A. A multi-line function
- B. An anonymous, single-expression function
- C. A replacement for classes
- D. Used only in AI

13. **What is encapsulation in OOP?** 1 point

Mark only one oval.

- A. Combining multiple classes
- B. Hiding internal details and exposing behavior
- C. Using inheritance
- D. Overloading operators

14. **What is the main difference between __str__ and __repr__?** 1 point

Mark only one oval.

- A. repr is slower
- B. str is for users, repr is for developers/debugging
- C. str replaces print()
- D. They behave exactly the same

15. **Why is this code useful?** 1 point
if hasattr(acc, "apply_interest"): acc.apply_interest()

Mark only one oval.

- A. Prevents syntax errors
- B. Ensures method exists before calling it
- C. Improves performance
- D. Makes code shorter

16. **Which is a good AI development best practice?** 1 point

Mark only one oval.

- A. Trust model output blindly
- B. Use one AI model for all tasks
- C. Choose models based on strengths and use cases
- D. Avoid documentation

17. What is the main purpose of context in an AI prompt? 1 point

Mark only one oval.

- A. To make the AI respond randomly
- B. To provide relevant information that helps the AI generate a better answer
- C. To increase the size of the AI model
- D. To replace the AI model

18. What is the main purpose of RAG? 1 point

Mark only one oval.

- A. To train an AI model from scratch
- B. To retrieve relevant information and use it to generate an answer
- C. To make AI responses shorter
- D. To remove information from a prompt